

TrES-5b

R-0964

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_230725 · created 2026-07-29T16:32:19+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460150.8652 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.013
Transit depth (Rp/Rs)^2	3.42 +/- 0.49 [%]
Orbital Inclination (inc)	85.61 +/- 1.78
Airmass coefficient 1 (a1)	1.033 +/- 0.027
Airmass coefficient 2 (a2)	-0.023 +/- 0.068
Scatter in the residuals of the lightcurve fit is	2.67 %
Best Comparison Star	#3 - [109, 239]
Optimal Aperture	2.8
Optimal Annulus	7.59
Transit Duration (day)	0.0819 +/- 0.0072

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.185 ± 0.013 vs 0.142 (deviation 2.45σ)
Duration fit vs published	0.0819 d vs 0.0758 d (deviation 0.63σ)
Mid-time O–C	-4.7 min
Depth SNR	10.6
Residual scatter	2.67 %

Light curve

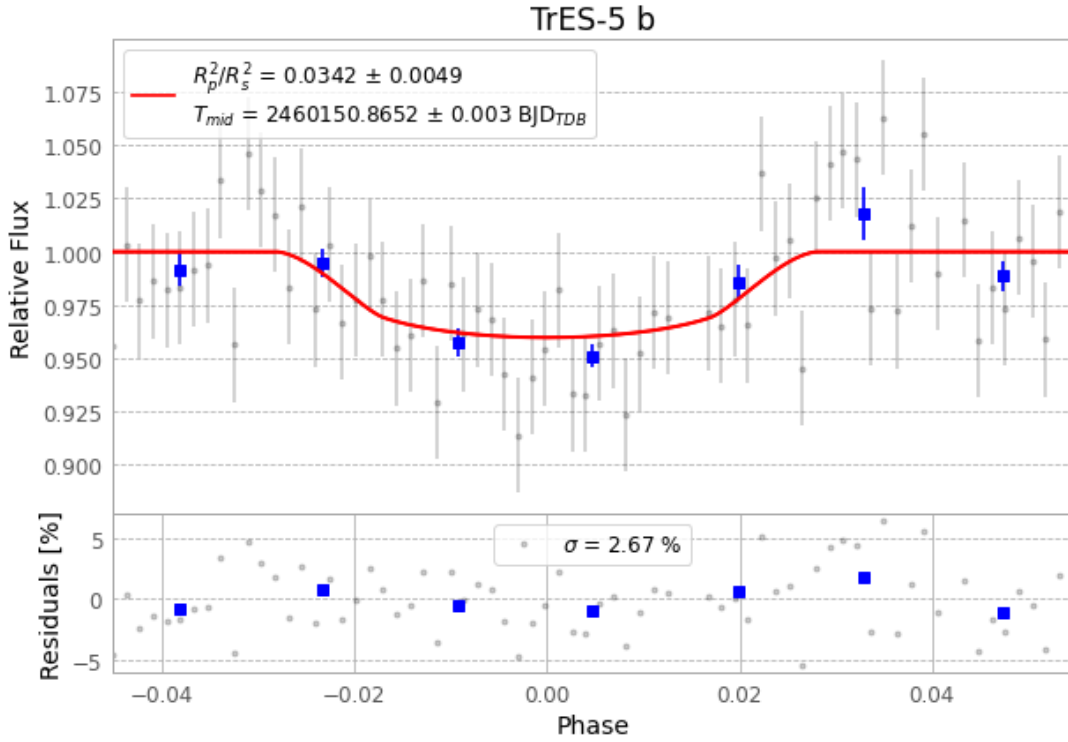


Figure 1 — FinalLightCurve_TrES-5 b_2023-07-25.png (SHA-256 5c684977a9ea6ce0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [248, 200], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 def5b8ca2bff824c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

72 FITS frames (47 MB), TRES-5230725070617.FITS → TRES-5230725103915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-230725.log (45a29dd7001b...), FinalLightCurve_TrES-5 b_2023-07-25.png (5c684977a9ea...), FinalLightCurve_TrES-5 b_2023-07-25.csv (ff5a5937797a...)

Compute resources

Wall time 240.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 16:32Z.